WELLSENSE · PROJECT PLAN

WellSense project plan

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/wellsense		
Source	docs/11-plan.md @ e3c296b		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-25	Amish Chadha	First issue (kit 1.9.1 upgrade) at stage P3

WellSense project plan

WellSense is at stage P3 (design outputs, TRL 3): the design is constructable, drawn, priced and planned for a prototype build, and every concept option is modelled at detail level D3. The next gate is Amish's concept decision between options A and C (WLS-DDR-005), with P4 (verification, TRL 4) on hold while the portfolio cap is P3. Method: STANDARDS sections 19 and 23.

Workstreams

Table 1. Workstreams.

Workstream	Owner (role)	Deliverable	Status
Problem and users	Project lead	WLS-PRB-001 with users and stakeholder tables	Done (v0.7); co-design not started
STEM check	Design lead	WLS-STM-001 and `docs/04-calcs/stem_check.py`	Open: S7, S10 and S21 need TRL 4 data; fixes D-S1 to D-S4 made
Concept options	Design lead	WLS-OPT-001, `cad/src/model.py` options A to C at D3	Done; choice awaiting Amish (WLS-DDR-005)
Requirements and traceability	Design lead	WLS-URS-001, WLS-PRS-001, WLS-RTM-001	Done; targets to revise in co-design
Calculations	Design lead	WLS-CAL-001 v0.6	Done for TRL 3
CAD and drawings	Design lead	`cad/src/model.py`, `cad/fidelity.json` (D3), STEP, WLS-DWG-001	Done
Build plan	Design lead	WLS-BLD-001	Done (plan only; no build at TRL 3)
Regulatory, feasibility, risk, market	Project lead	WLS-REG-001, WLS-FEA-001, WLS-RSK-001, WLS-MKT-001	Done (first issue); certificates to assess
Electronics interface	Electronics lead	Interface board from bought modules; FieldNode pinout signed off (WLS-DEC-001)	Designed on paper; no PCB at TRL 3
Firmware and dashboard	Software lead	FieldNode sensor driver, conversion, dashboard (PR-013)	Not started (TRL 4 and later; firmware only as a labelled sketch before then)
Partner and co-design	Project lead	Approach ACWADAM; co-design sessions	Not started
Renders and storefront	Coordinator (studio machine)	`media/options.png` (Cycles), product renders from `cad/src/product_model.py`, cards	Awaiting the coordinator's render run

Resources needed

Table 2. Resources (estimates; no money is committed while the portfolio is capped at TRL 3).

Resource	Deliverable	Effort	Estimated cost (USD)
Co-design partner time (field team)	Well sizes, site access, co-design sessions, dashboard language	3 to 5 days	To agree with the partner
Prototype parts, one logger	Build to WLS-BLD-001 (TRL 4, on hold)	Parts only	264.40 at 30 m plus 139.00 for the FieldNode core (BOM)
Shared electric tape meter	Datum, calibration and checks (BOM line 11)	One per partner	80.00 (BOM estimate)
Workshop time	Build and bench check (TRL 4, on hold)	About 2 days for two people	Volunteer or partner
Drinking-water certificates	Supplier documents for probe, cable, HDPE, EPDM (PR-009)	Supplier enquiries	To assess
Test laboratory or university	Side-by-side accuracy and drift record (PR-004, PR-005)	3 to 12 months of field record	To assess

Stage gates

Table 3. Stages (P4 to P6 on hold while the stage cap is P3).

Stage	Gate criteria	Status
P0 Proposal	PRB with users and stakeholders, STM, OPT, REG, FEA, PLN	Met
P1 Definition	URS, PRS draft, RSK with hazard analysis, MKT	Met
P2 Design inputs	PRS with targets and verification, CAL, RTM, concept selection DDR, STEM check closed	Met except the STEM check, which stays open on S7, S10 and S21 (TRL 4 data) and the concept choice (awaiting Amish)
P3 Design outputs	Model, STEP, DWG, priced BOM, BLD, DEC, design FMEA, traceability complete	Met (2026-10-25, kit 1.9.1)
P4 Verification	Bench and field tests (TST), verification report	On hold (TRL cap 3)
P5 Validation	Field trial with the partner, certification status	On hold
P6 Commercialization	Launch with a named deployment partner	On hold

Next steps and decision requested

1. **Decision requested of Amish (O-1):** choose the concept to carry forward, option A (current design) or option C (acoustic sounder), which scored within 10 % (173 against 164; WLS-OPT-001, WLS-DDR-005). Recommendation: none, as the rule asks; the repo stays on A meanwhile.

2. Review the four fixes made under the standing instruction (WLS-DEC-001 D-S1 clevis pin, D-S2 drift test, D-S3 span from the probe's certificate, D-S4 loop fuse); any can be reversed.
3. Approach ACWADAM (decided 2026-10-02) and ask for well sizes and a site with a pump.
4. Ask suppliers for drinking-water certificates (PR-009) and a quote for a 22 mm, 0.25 % class vented probe.
5. When the cap rises: TRL 4 bench build and a side-by-side tape record (PR-004, PR-005), and an installation trial beside a pump (PR-010, PR-011).

References

Every source cited in this repo's documents, Chicago style.

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- Standards Council of Canada. "NSF/ANSI/CAN 61 - 2023." https://scc-ccn.ca/standardsdb/standards/4032142.
- UNESCO. *The United Nations World Water Development Report 2022: Groundwater: Making the Invisible Visible.* Paris: UNESCO, 2022. https://www.unesco.org/reports/wwdr/2022/en.
- World Health Organization. *Guidelines for Drinking-Water Quality: Fourth Edition Incorporating the First and Second Addenda.* Geneva: WHO, 2022. https://www.who.int/publications/i/item/9789240045064.
- Standards and practice references named in WLS-STM-001 and WLS-REG-001: AWWA C654-13 (disinfection of wells), ASTM D1785 (PVC pipe), IEC 60529 (IP code), IEC 61140 (protection against electric shock), ISO 4373:2022 (hydrometry, water level measuring devices), HSE INDG143 (manual handling), Texas Instruments ADS1115 datasheet, Semtech AN1200.13 (LoRa time on air).

- Market benchmarks in WLS-MKT-001: Instrumart (HOBO U20L), Fondriest (Solinst Levelogger 6), Acorn Controls on IndiaMART (DWLR), all checked 2026-10-25.